

The decision to build the Space Station

Too weak a commitment?

Howard E. McCurdy

The decision to build the Space Station illustrates the way in which the incremental nature of policy making in the US government favours short-term flexibility over long-term commitments. In making the moves necessary to win approval for the Space Station, NASA officials and their allies took actions which gave policy makers frequent opportunities to re-examine the programme once approved, and may have made likely the continuing uncertainty over the future of the station. Difficulty in securing long-term commitments for large and complex science and technology programmes has been a central feature of US space policy ever since the landing on the Moon.

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In January 1984, President Ronald Reagan directed the National Aeronautics and Space Administration 'to develop a permanently manned space station and to do it within a decade'.¹ In the financial plan for the Space Station programme approved within the White House at that time, NASA officials received permission to request at least \$2.2 billion for fiscal year 1988 in order to build the facility.² In fact, the US Congress appropriated a mere \$425 million for the station in that year. Following smaller appropriations in previous years, this level of funding put the Space Station at least temporarily on hold – one more incident in the programme's continuing search for survival. Almost from the moment it was first approved, the Space Station programme has been rescued time after time, only to find itself in jeopardy anew.

The problems of the Space Station programme could be blamed on bad luck, bad timing, and a seemingly immutable federal budget deficit which discouraged Congress from funding new initiatives of any sort. Rather than view the fits and starts of the programme as a temporary aberration in the relentless march of science and technology, however, I will argue in this article that the Space Station's agonies are the natural result of the political environment within which the US space programme has had to operate since the lunar landing in 1969. In adjusting to that environment, NASA officials have learned to accept programme approval under conditions which make long-term commitments difficult to achieve. This has created a significant dilemma for advocates of major new space initiatives – whether to pursue approval for programmes such as the Space Station under conditions which erode the long-term stability necessary to complete them or any complex scientific or technical project, or not to pursue the programmes at all. This dilemma, moreover, has come to affect all of NASA's major space programmes, from the Space Shuttle to current efforts to reach towards Mars.

The Space Station decision

At the time, President Reagan's directive in his 1984 State of the Union

¹Weekly Compilation of Presidential Documents, 25 January 1984, p 87.

²Letter to Bart Borrasca, OMB, from Peggy Finarelli, NASA, 8 September 1983. NASA, *Budget Estimates, Fiscal Year 1985*, Volume I, Summary Data (Office of Management and Budget Submission), p SUM 9.

message seemed to conclude a two-year debate within the administration over whether or not the USA should start work on a permanently occupied station in space. In making his decision, Reagan overrode the advice of many of his principal advisers, including his Secretary of Defense, his budget director, and his chief science adviser, all of whom opposed the station initiative. Within NASA, the President's decision was viewed as a decisive victory in the agency's long-standing quest for a human outpost in low-Earth orbit.

In retrospect, the decision has turned out to be less decisive than it seemed. The history of events leading up to the President's decision provides the backdrop for an analysis of why the advocates of the Space Station won approval to start the programme, but without a strong commitment to finish it.

'Establishment of a permanent near-Earth space station' has been a major goal within NASA ever since the US space agency was created. This was a natural result of the historic commitment to human flight and space exploration within several of the organizations welded together to form NASA. In a long-range plan issued in late 1959, NASA officials predicted that the Space Station would be built prior to any attempt to land humans on the Moon.³ President John F. Kennedy bypassed this particular approach when he learned in 1961 that the USA in all probability could not beat the USSR in putting 'a multi-manned laboratory into space'.⁴

In the euphoria of the 1969 lunar landing, NASA officials proposed the construction of a 12-person station followed quickly by a 50 to 100-person Earth-orbiting 'space base'.⁵ As envisioned by its advocates, elements of these frontier stations would be delivered into orbit by a two-stage derivative of the giant Saturn V. Simultaneously, NASA wanted to develop a three-pronged space transportation system, one element of which would consist of a 'chemically fueled shuttle' designed to ferry people and equipment between the Earth and Space Station 'in an airplane-type mode'.⁶ When the Apollo 11 celebration died down, space buffs discovered that they lacked political support for anything more than one new initiative. Responding to this realization, NASA officials adopted an incremental strategy. They decided to take the first step in the space transportation system (the Space Shuttle), and pursue the Space Station only once the Shuttle became operational.

Among top NASA officials, the Skylab programme was not viewed as a substitute for the Space Station goal. Skylab – an orbital workshop – was an outgrowth of the Apollo programme. It did not meet NASA's basic definition of a Space Station, being neither permanently manned nor – as it tumbled towards Australia – permanently in orbit.⁷

In mid-1981, when NASA Administrator-designee James M. Beggs and Deputy Administrator-designee Hans Mark announced that the agency would try once again to win approval for a permanently occupied space base, the Space Station was a non-issue on the US political agenda. Only a few people in Washington were aware of NASA's 23-year-old enthusiasm for a Space Station. On the basis that NASA had no new manned space flight initiatives worth offering, budget director David Stockman proposed to pare NASA spending from \$6.3 to \$5.2 billion by FY 1986.⁸

After Beggs and Mark announced that NASA would seek presidential approval for the Space Station, the initiative encountered strong opposition. Stockman called NASA's proposal for a multi-billion dollar

³NASA, Office of Program Planning and Evaluation, *The Long Range Plan of the National Aeronautics and Space Administration*, 16 December 1959.

⁴Office of the Vice President, Memorandum for the President, Subject: Evaluation of Space Program, 28 April 1961, p 3. A complete description of these events is contained in John Logsdon, *The Decision to Go to the Moon*, MIT Press, Cambridge, MA, 1970.

⁵NASA, *The Post-Apollo Space Program: Directions for the Future*, September 1969. The proposal was endorsed by the Space Task Group, Report to the President, *The Post-Apollo Space Program: Directions for the Future*, September 1969.

⁶Space Task Group, *The Post-Apollo Program*, *op cit*, Ref 5, p 15.

⁷W. David Compton and Charles D. Brown, *Living and Working in Space: A History of Skylab*, SP 4208, NASA, Washington, DC, 1983.

⁸US Executive Office of the President, *America's New Beginning: A Program for Economic Recovery*, 18 February 1981, p 6-36.

Space Station an example of 'high-tech socialism' which the nation could ill afford.⁹ 'Why should we put men in space,' presidential science adviser George Keyworth asked, 'when we are just entering the era of robots and automation?'¹⁰ A special Defense Department study group, led by Air Force officials, concluded that a manned space station would provide no significant improvement over conventional means of conducting national security operations in space. Defense Secretary Caspar Weinberger personally lobbied against programme approval. 'My reservations,' he wrote, 'relate to cost and impact on the Space Transportation System.' Approval of a Space Station would take funds from defence programmes and 'would inevitably divert NASA managerial talent and resources from the priority task of making the Space Transportation System fully operational', Weinberger concluded.¹¹

The absence of an approved long-range plan for the exploration of space seriously complicated NASA's efforts to develop a rationale for the Space Station and to build a constituency for it. Since the founding of the agency, NASA officials had viewed the Space Station as a facility necessary for the long-range exploration of space by humans. Lacking an approved long-range objective that might justify a Space Station (such as a human interplanetary voyage), NASA officials turned to missions (such as space manufacturing) which might be performed from a Space Station in the 1990s. Many of those missions, however, could be performed less expensively by machines. When George Keyworth urged NASA to 'let the American people share the grand vision of the future of space' by proposing a lunar base or an expedition to Mars, NASA officials treated his suggestion as a red herring.¹² Such a proposal, NASA officials stated, would simply divert attention from winning approval for the objective at hand.

In order to think through the project and organize NASA's case for the new initiative, Beggs established a special Space Station Task Force. Announcement of the Task Force was made in May 1982, six weeks prior to the fourth and final test flight of the Space Shuttle *Columbia*, when NASA officials expected the Shuttle to 'go operational'. Not only did Task Force members have to develop a convincing Space Station rationale for outside consumption, they also had to unify thinking within the agency, where 23 years of Space Station studies had left a wide diversity of opinion on how best to proceed. In its two years of existence, the Space Station Task Force grew to 72 full-time members and involved over 500 NASA employees in its working groups and steering committees. The Task Force leaders – John Hodge, Robert Freitag, Terry Finn and Daniel Herman, along with Philip Culbertson in the Office of the Administrator – formulated many of the strategies which resulted in approval for the Space Station.

Advocates of an aggressive space programme found in Ronald Reagan their most ardent presidential supporter since John F. Kennedy in 1961. In spite of his general enthusiasm for space, however, Reagan refused to endorse the Space Station outright. NASA officials and their allies were thus obliged to put the station initiative into the White House policy review labyrinth. They struggled to push the Space Station plans through the White House budget review process. When that failed, they put the issue into the Senior Interagency Group for Space, an arm of the National Security Council. When they could not get a favourable recommendation out of that group, they tried the Cabinet Council on Commerce and Trade. In probing for White House support, NASA

⁹David A. Stockman, *The Triumph of Politics*, Avon Books, New York, 1986, p 43 of photo section.

¹⁰Shirley Hobbs Scheibla, 'Magic Formula?' (transcript of an interview with George Keyworth), *Barron's*, Vol 62, 6 December 1982, p 9.

¹¹Letter to James M. Beggs, NASA Administrator, 16 January 1984, from Caspar Weinberger, Secretary of Defense.

¹²George A. Keyworth, 'The U.S. space program – where do we go from here?', proposed remarks to the 19th Joint Propulsion Conference, Seattle, WA, 27 June 1983, p 14.

officials sought to avoid negotiations of the sort that had compromised the Shuttle programme during its approval process 12 years earlier.¹³ Advocates of the Space Station sensed that President Reagan would approve the programme if only they could put an uncompromised vision of it in front of him. 'It was clear to me that the President favored the idea,' NASA Administrator James Beggs explained. 'I knew I had the only vote that counted.'¹⁴

After obtaining the presidential directive approving the Space Station, NASA officials encountered significant opposition on Capitol Hill. Resistance was especially strong in the House Appropriations Subcommittee which handled the agency's budget.¹⁵ Much of the opposition flowed from scepticism about NASA's estimates of the cost of the facility. 'I will bet you a hat,' said Edward Boland (Democrat, MA), the House Subcommittee chair, 'that it is going to cost a lot more than \$8 billion' – NASA's announced cost goal.¹⁶ Boland asked NASA officials to consider building what became known as a 'man-tended' Space Station, one that would be visited periodically by astronauts rather than start off as 'permanently manned'. While Boland and his allies did not have the votes to force NASA to accept a man-tended facility, NASA officials and their allies also did not have the votes to win a clear Congressional commitment to build a permanently manned version. NASA began work on the programme in FY 1985 with a \$155.5 million budget which did little more than allow the agency to undertake phase B definition and preliminary design activities to study the Space Station, not to build it.

In winning approval for the Space Station, NASA officials encountered many of the forces that characterize incremental policy making: a lack of consensus about objectives, the preference of policy makers to review new initiatives one at a time on the basis of established programmes, the reluctance of policy makers to make commitments which might limit future options, opposition from other agencies to proposals which would significantly increase NASA's share of the federal budget, and a tendency to define 'rational' policy as that on which participants could agree rather than that which best met long-term goals.¹⁷ NASA officials and their allies have laboured within that sort of political environment ever since landing the first humans on the Moon. With the Shuttle decision, NASA officials were forced to seek approval for new starts within the normal world of incremental politics. The Space Station decision in many ways represents NASA's maturation in the incremental game.

The triumph of incrementalism

The Apollo decision provided a rare departure from the norms of incremental decision making which tend to dominate US government. President Kennedy selected a single, unambiguous goal from a list of possible alternatives. NASA officials were given plenty of discretion to select the most appropriate method for attaining that goal, and chose lunar rendezvous. The agency received resources sufficient to accomplish the goal, and the criterion for determining the success or failure of the mission was clearcut.¹⁸

The history of the Space Station programme, on the other hand, has more closely followed the model of incremental decision making. With an incremental decision, policy makers begin from an established base

¹³John Logsdon has written extensively on the approval of the Space Shuttle programme. See 'The decision to develop the Space Shuttle', *Space Policy*, Vol 2, May 1986, pp 103–119; 'The Space Shuttle decision: technology and political choice', *Journal of Contemporary Business*, Vol 7, No 3, 1978, pp 12–30; and 'The Space Shuttle program: a policy failure?', *Science*, No 232, 30 May 1986, pp 1099–1105.

¹⁴James Beggs, interview conducted by Howard McCurdy, 3 September 1985.

¹⁵Hans Mark, *The Space Station: A Personal Journey*, Duke University Press, Durham, NC, 1987, Ch 16.

¹⁶US House Subcommittee of the Committee on Appropriations, *Department of Housing and Urban Development—Independent Agencies Appropriations for 1985*, part 6, 98th Congress, Second Session, p 2.

¹⁷For a description of the elements of incrementalism, see Charles Lindblom, 'The science of "Muddling Through"', *Public Administration Review*, Vol 19, spring 1959, pp 79–88. See also Aaron Wildavsky, *The New Politics of the Budgetary Process*, Scott, Foresman, Glenview, IL, 1988.

¹⁸Logsdon, *op cit*, Ref 4. Courtney G. Brooks, James M. Grimwood and Loyd S. Swenson, *Chariots for Apollo: A History of Manned Lunar Spacecraft*, SP-4205, NASA, Washington, DC, 1979.

(generally defined as what the agency did last year) and direct their attention not to the programme as a whole but to small increments within it. This approach to policy making places a premium on flexibility – the ability of policy makers to move forward or backward from the status quo as conditions change. Policy makers using the incremental approach are typically reluctant to embrace a single, long-term objective, since this would limit their flexibility to make mid-course corrections as experience accumulates. Because goals remain general rather than specific, year-to-year decisions are typically settled through bargaining among the participants rather than the sort of ends-means analysis which characterizes comprehensive review.

The Apollo decision has achieved something of a mythical quality within the US space programme. Many persons have argued that NASA works best when pursuing a unifying policy such as that created by the lunar goal.¹⁹ For nearly 20 years, however, NASA officials have been obliged to operate within a political system which treats Apollo-type decisions as anomalies. That same system prizes flexibility and incremental review. In seeking approval for new initiatives in the post-Apollo era, NASA officials have learned to adopt strategies which emphasize flexibility over commitment as a price for getting new programmes approved.

The Space Station decision illustrates well how flexibility gets substituted for commitment in the US space programme. In presenting the Space Station for political review, NASA officials had to formulate answers to key conceptual questions. What would a Space Station do? What would it look like? And what would it cost? The answers which NASA officials provided to these questions tended to make it easier for policy makers to modify the programme once approved, in spite of the desire of agency officials to obtain long-term commitments. This happened in a natural sort of way, based on lessons learned from previous programmes, especially the decision to approve the Space Shuttle.

Emphasize missions, not configurations

In one of their first major moves after starting up the Space Station Task Force, in 1982, John Hodge and other Task Force leaders decided to identify the missions to be performed on the orbital facility before starting any design work. 'On the Task Force,' Hodge explained, 'people got very mad at me, because there were a lot of engineers on the Task Force, mostly engineers, and they want to start designing things and drawing pictures.'²⁰ Instead, Hodge set up a working group which identified 107 missions to be performed on the Space Station between 1991 and the year 2000, using studies prepared by potential users and aerospace contractors.²¹

Hodge was well aware of an internal NASA study, conducted by Langley Center Director Donald P. Hearth, which criticized NASA's tendency to plunge into design details before consulting with users about the purpose of a project.²² Hodge decided that NASA would spend up to 10% of the total programme cost and at least two years *after* the Space Station was approved defining the requirements of the facility and developing a baseline design.

Beyond the findings of the Hearth report, NASA's desire to avoid configuration debates during the approval process was also based on

¹⁹Report of the Transition Team, George M. Low, Team Leader, to Mr Richard Fairbanks, Director, Transition Resources and Development Group, 19 December 1980. See also Sally K. Ride, *Leadership and America's Future in Space*, a report to the Administrator, NASA, August 1987, and US National Commission on Space, *Pioneering the Space Frontier*, Bantam Books, New York, 1986.

²⁰John Hodge, interview conducted by Howard McCurdy, 27 June 1986.

²¹NASA, Space Station Task Force, *Space Station Program Description Document: Book 2, Mission Description Document*, March 1984.

²²Donald P. Hearth, NASA Project Management Study, 21 January 1981.

agency experience with getting the Shuttle approved. During the White House review of the Space Transportation System in 1971 and 1972, the actual design of the Space Shuttle became an item for negotiation. 'Everybody was a shuttle designer,' moaned one participant.²³ The fact that economists in the Office of Management and Budget would even consider designing a high-performance spacecraft horrified NASA officials. Nonetheless, in order to win support within the White House for the programme, top NASA officials felt obliged to negotiate design details such as the size of the Shuttle payload bay.

Hodge's strategy effectively precluded that possibility. By giving policy makers no pictures to redraw, he removed the design of the Space Station as a bargaining chip in the approval process. Opponents could not criticize its design if NASA published no design. In adopting this strategy, NASA officials substituted flexibility for stability. By publishing no design, however, NASA officials committed themselves to an extensive post-approval definition and design period. This guaranteed that the preliminary design of the Space Station would not be set until at least 1985 and that the final design would not be frozen until 1989 – five years after the programme was approved. This gave opponents and other officials in the White House and the Congress five more years to chip away at the design, costs and goals of the Space Station; five years during which enthusiasm for the programme would rise and fall.

Distributed architecture

Because they were adding up missions, rather than selecting among competing designs, leaders of the Task Force were able to include as part of the Space Station programme objectives which might prove incompatible on a single, fully developed facility. As is typical of incremental policy review, proponents of the Space Station adopted general objectives and did not become specific about programme details until potential users had agreed to support the programme.

Task Force members knew that the Space Station would have to appeal to a large constituency, if for no other reason than to counter the breadth of the opposition. NASA officials sought to include within their constituency people entranced by the potential for manufacturing in space; people who wanted a service station from which to repair satellites and transit to geosynchronous orbit; people who wanted to engage in space science, astronomy and Earth observation; national security and arms control officials who needed orbital reconnaissance platforms; international partners who saw the advantages of sharing in US space technology; and space advocates who wanted to explore and eventually settle the Solar System. In identifying the missions to be performed on the Space Station, Task Force officials appealed to all of these groups. Included within the list of 107 missions were a satellite servicing centre, an orbital transfer vehicle, a large deployable reflector, an Earth science research project, a materials processing laboratory, a Mars sample return mission, and more.

Task Force leaders understood that these functions could not be performed from a single, multi-functional facility. They therefore designed more than one. They called it 'distributed architecture', a concept developed by Task Force member Daniel Herman. What this meant was that the 'Space Station programme' would produce not only a live-in modular core with associated laboratories but also two

²³Quoted in Logsdon, 'The Space Shuttle program', *op cit*, Ref 13, p 1101.

unmanned platforms, one in the same 28.5 degree orbit as the live-in core, the other circling the poles. Including the unmanned platforms in the overall Space Station programme helped to meet the various needs of the science community.

Making the Space Station attractive to international participants was more complicated. The potential for international cooperation on the Space Station with Europeans, Canadians and Japanese was a major selling point – as important to many law-makers on Capitol Hill as the role of the Space Station in promoting space commercialization was to Reagan Republicans in the White House. For domestic political reasons, however, international partners could not participate in a US Space Station on which the Defense Department was conducting weapons research.

International partners also insisted upon the principle of mutual access, which would give them the right to use each facility on the Space Station regardless of who built it. And the Europeans expressed interest in building a laboratory module which would duplicate US operations on the Space Station such as materials processing, and would be constructed in such a way that it could be detached from the station core for independent operation.

The national security issue temporarily disappeared in 1983 when officials in the US Defense Department announced that they saw no need for a Space Station and refused to participate in its development. As for other sticky international issues, NASA officials made a typically incremental suggestion. Let us agree in general to cooperate, they said, and put off our differences until we have developed a preliminary design for the facility.

That left the potential conflict between space commercialization and space expedition on the live-in core. The promise of microgravity industries in space required laboratory facilities which remained as still as physically possible. The use of a Space Station as a base for deep-space operations, however, required a facility which would move around as spacecraft docked and fuel storage tanks were filled and drawn down. To the extent that the Space Station would serve as a test vehicle for interplanetary spacecraft, it might even require artificial gravity.

NASA officials did not emphasize these inconsistencies during the approval process, preferring instead to let it appear as if a Space Station programme could accomplish all of these functions. 'Space Station was carefully crafted and described as anything you want it to be,' one official observed.²⁴ During the approval process, NASA officials did not discuss openly the possibility that full-scale development of these functions might require what they termed 'branching', the construction of two or more separate Space Stations. The last thing that either opponents or supporters of the programme wanted to hear was that NASA wanted to build two Space Stations rather than one.²⁵

At some point in the design process, potential inconsistencies such as these had to be resolved. NASA officials correctly concluded, given the nature of US government, that the programme approval process would be a poor place to resolve them. Premature discussion of programme details, before influential parties had agreed to support the programme, would divide NASA's supporters. By putting those issues off, of course, NASA officials built an extra measure of flexibility into an already loose post-approval process.

²⁴This quotation is taken from an interview transcript in the author's possession.

²⁵The branching concept was not discussed openly within NASA until one year after the programme was approved. NASA Headquarters, Office of Space Station, *Proceedings of the Space Station Evolution Workshop*, Williamsburg, VA, 10–13 September 1985.

Defining programme costs

No aspect of the Space Station decision better exemplifies the incremental nature of the process than the effort to define programme costs. Initially, NASA officials refused to set a specific cost figure for the Space Station. 'Space stations,' James Beggs announced, 'are the kind of development that you can buy by the yard.'²⁶ By this he meant that NASA planned to start with a modest facility and add more modules as money became available. Task Force leaders argued that it would be premature to set a specific cost figure until the agency completed its phase B studies, well after the programme was approved.

NASA's reluctance to be specific about costs fuelled the arguments of the opposition. Opponents suggested that NASA executives did not know total programme cost or – worse still – that the costs were known but astronomical. In fact, NASA officials had established in April 1983 a Concept Development Group under the Space Station Task Force to study various station requirements and the ranges of possible costs related to meeting them.

In summer 1983, as the Space Station programme moved towards a presidential decision, Beggs committed NASA to an \$8 billion cost figure. Beggs selected \$8 billion because it was the highest number that he could justify within the anticipated NASA budget over the next few years. 'If you picked a number too high,' Beggs explained, 'OMB would come back and argue that we were busting the budget in the out years. And if you picked too low a number you know you would be in the soup very quickly.'²⁷ Task Force Director John Hodge remembered how people outside NASA reacted to the various cost bands that his group had examined. 'I reached the scream level at about \$9 billion,' Hodge recalled.²⁸

NASA's Space Station programme now had to accommodate both the \$8 billion figure and the first 33 missions to be conducted on it. This was only possible if NASA officials issued a very narrow definition of what the \$8 billion included. Space agency planners had done this in the past. The much-heralded \$20 to \$40 billion cost goal for the Apollo programme included only the first landing on the Moon, not the five which followed. The \$5.15 billion cost goal for the Space Transportation System took the programme through the test phase only, and included just two of the five orbiters NASA officials wanted to buy.

NASA's definition of the \$8 billion cost goal for the Space Station did not include the money needed to transport the facility to orbit. It did not include the cost of a 'lifeboat' which the crew could use to return to Earth in an emergency. It did not include the cost of operating the station or conducting experiments, nor the cost of future additions. And it included only a small reserve for technological uncertainty.²⁹

The \$8 billion estimate covered little more than the cost of building the initial Space Station hardware. *That estimate was quite accurate.* The hardware estimate grew by only \$1 billion as a result of changes in the approach to the design and assembly of the Space Station worked out during the phase B studies.

In early 1987, the Office of Management and Budget (OMB) announced that the cost of the Space Station had 'grown' to \$14.5 billion – an 81% cost overrun prior to the construction of any major part of the facility.³⁰ In fact, nearly all of that growth consisted of additions to cost reserves, ground-based infrastructure, and technical support of the field

²⁶US Senate Subcommittee on Science, Technology and Space, Committee on Commerce, Science and Transportation, *NASA Authorization for Fiscal Year 1984*, 98th Congress, First Session, 9 March 1983, p 51.

²⁷James Beggs, interview conducted by Howard McCurdy, 30 October 1987.

²⁸John Hodge, interview conducted by Howard McCurdy, 10 July 1985.

²⁹Space Station Task Force Roundtable, a discussion with John Hodge, Robert Freitag, Daniel Herman and Terence Finn conducted by Howard McCurdy, Sylvia Fries and Adam Gruen, 4 August 1987.

³⁰James C. Miller, Director, Office of Management and Budget, 'Memorandum for the President, subject: revised cost estimates for the Space Station', 10 February 1987.

centres – costs which the leaders of the Space Station Task Force and Concept Development Group had defined out of their estimate.

Faced with an OMB ultimatum and a restive Congress, NASA Administrator James Fletcher decided in early 1987 to scale back the Space Station to fit the expanded definition of programme costs. He did it 'by the yard', cutting out (NASA officials preferred the term 'deferring') one of the two co-orbiting platforms, the satellite servicing bay, and the two vertical keels on the central facility and their instrument attachment points.³¹

By refusing to go forth in 1983 with a detailed financial plan, NASA officials managed to avoid a White House staff judgement on the cost and configuration of the Space Station. They thus avoided the bargaining which had trapped their predecessors in seeking approval for the Space Shuttle. In negotiating the costs and configuration of the Space Transportation System, NASA officials had bargained themselves into a Shuttle configuration which many believed was technologically inferior to the two-stage reusable system and not cost-effective.³²

NASA officials were determined to avoid, as a condition for gaining President Reagan's approval of the programme, Shuttle-type negotiations with opponents of the Space Station in the OMB. To do that, agency officials had to adopt an incremental attitude toward programme costs which allowed policy makers to adjust annual appropriations as political support for the programme oscillated.

Future programmes

As a result of strategies formulated during the approval process, NASA officials presented policy makers with a Space Station concept which allowed people outside of NASA to exercise a great deal of discretion over the programme as it developed. The purpose, design and cost of the Space Station were subject to extensive review after the programme had been approved. NASA officials created a programme which lacked the sort of stability that had characterized the Apollo era. However, they succeeded in gaining approval for the Space Station, a policy success which few people at that time thought the agency could achieve.

In order to preserve support for the Space Station programme as approved, NASA officials had to count on a succession of events which in fact have not come to pass. They had to maintain the technological momentum created by the initial success of the Space Shuttle. They had to fit their appropriation requests into an agency budget caught between existing programme costs, including the Shuttle, and the modest 1% real growth in the NASA budget which President Reagan had approved in December 1983. And the government as a whole had to avoid the sort of fiscal crises which would cause policy makers and heads of other agencies to attack any agency's programmes, particularly major new starts.

NASA officials achieved programme approval, but without the level of political commitment that would allow the Space Station to weather the bad times which have followed. Had NASA executives insisted upon a stronger long-term commitment to the programme in 1984, ignoring the political demands for flexibility, the Space Station probably would not have been approved. As NASA Administrator James Fletcher observed, 'There simply is no way, in our system of government, to get a long-term national commitment.'³³

³¹'NASA proceeding toward Space Station development', *NASA News*, release 87-50, 3 April 1987.

³²Logsdon, 'The decision to develop the Space Shuttle', *op cit*, Ref 13.

³³James C. Fletcher, 'Space Shuttle development', *Science*, 18 July 1986, 'Letters', p 263.

Table 1. Presidential requests for first-year appropriations for major new initiatives (millions of dollars).

Flight to the Moon (FY 1962)	\$531 (1961 dollars)
Space Transportation System (FY 1973)	\$200 (1972 dollars)
Space Station (FY 1985)	\$150 (1984 dollars)
Project Pathfinder (FY 1989)	\$100 (1988 dollars)

This phenomenon has come to affect the full spectrum of NASA activities, from human space flight to science programmes. Beyond the bureaucratic politics which shaped the Shuttle programme, fits and starts have characterized the Voyager programme, the Hubble Space Telescope and the Galileo probe.³⁴ The pattern appears to be repeating itself in the efforts to establish a long-term goal for the US space programme. Rather than insist upon a specific commitment to a mission to Mars, NASA officials have requested approval for Project Pathfinder, 'a systematic development of space technologies . . . which will aid us in deciding where this new adventure should take us.'³⁵

The extent to which NASA has been obliged to adopt incremental strategies in order to get programmes approved is revealed in Table 1. The table identifies the budgets approved by various presidents when four major initiatives were first approved. A large appropriation request tends to represent a strong national commitment; a small request indicates that policy makers may want to come back and modify the programme later.

In the agency's 30-year history, NASA officials have accepted less and less of a first-year budget commitment in order to get new programmes started. The trend is even more dramatic than it appears, since the figures are stated in the currency of the time, rather than in common-value dollars.

This tendency need not be viewed as a failure in policy making. In fact, from the point of view of people trying to get new initiatives approved against great odds, it represents something of a policy victory. What it does mean, however, is that new starts for major space efforts are not exempt from the same incremental forces that affect other government programmes. Even though NASA officials must carry out science and engineering programmes with very long lead times, they and their allies have to return year by year and rekindle the political support which brought the programmes approval in the first place. In moving towards new space missions such as an expedition to Mars, the US government will probably continue to take small incremental steps rather than giant leaps. People who yearn for dramatic, long-term commitments like the Apollo programme are likely to be disappointed for some time to come.

³⁴In addition to John Logsdon's work on the Space Shuttle decision, cited in Ref 13, see J. K. Davies, 'A brief history of the Voyager Project: the end of the beginning', part 1 of 6, *Spaceflight*, Vol 23, February 1981, pp 35-41; Paul Hanle, 'Astronomers, Congress, and the large space telescope', *Sky and Telescope*, April 1985, pp 300-305; and chapter 9 on the space telescope in Richard Barke, *Science, Technology, and Public Policy*, CQ Press, Washington, DC, 1986, pp 181-207. A study by Craig Waff on Project Galileo is currently underway within the NASA History Office.

³⁵*Weekly Compilation of Presidential Documents*, 25 January 1988, p 119.